

URBAN HEAT MITIGATION

AI / ML ANALYSIS SYSTEM

Geospatial Physics-Informed Analysis & Cooling Optimization

City: Delhi | 2026-08-21 02:18

24.6%

City Hotspot Area

+7.4°C

Urban Heat Excess

1.0000

PIML Model R²

0.47°C

Optimal Cooling

Framework:

Physics-Informed XGBoost + LSTM + SHAP + Getis-Ord Gi*

Data Sources:

Synthetic Landsat LST proxy, ERA5 atmospheric, OSM morphology

Grid:

200×200 cells @ 100m resolution

Interventions:

Urban Greening, Cool Roofs, Green Roofs, Water Bodies, Permeable Pavements, Street Trees

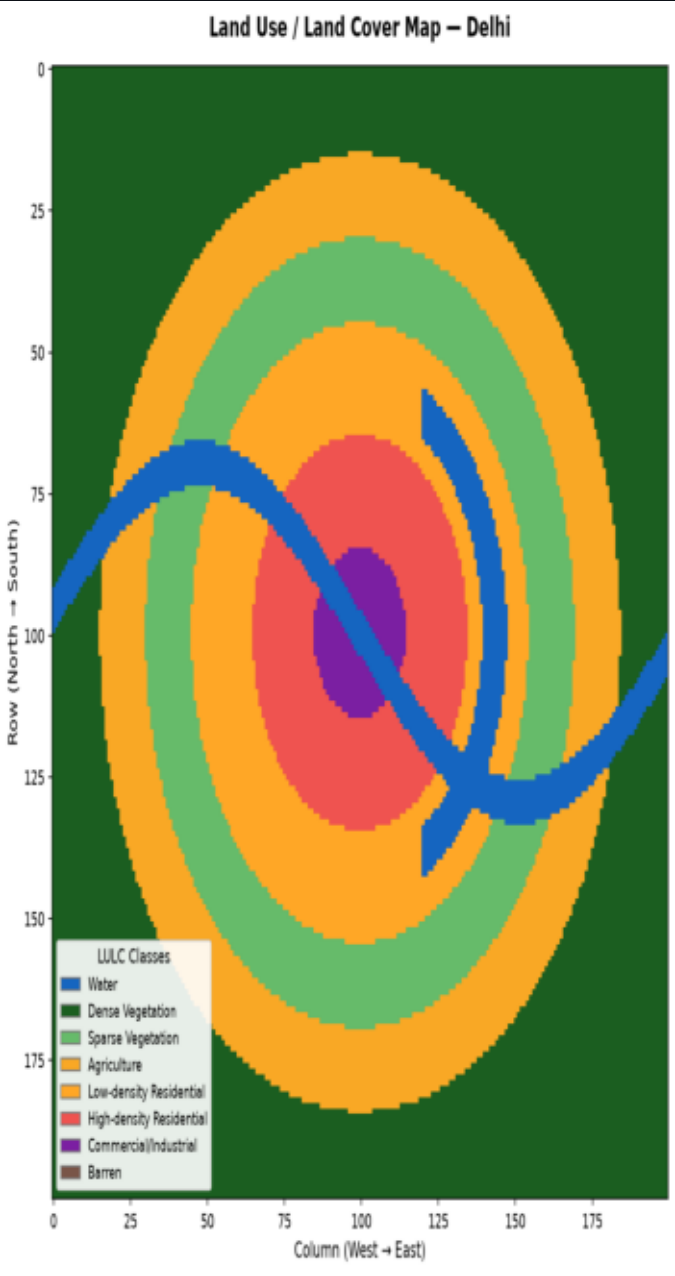
Optimizer:

Greedy budget-constrained, Budget=50,000 units

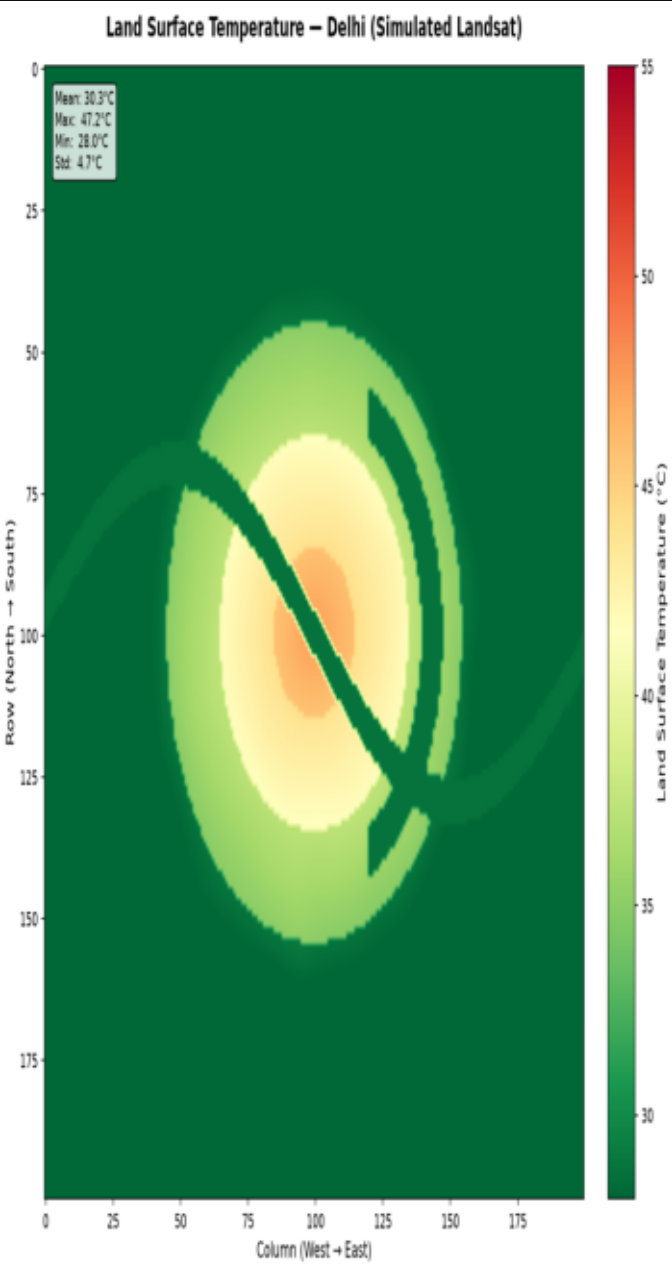
Urban Heat Stress Maps

City: Delhi

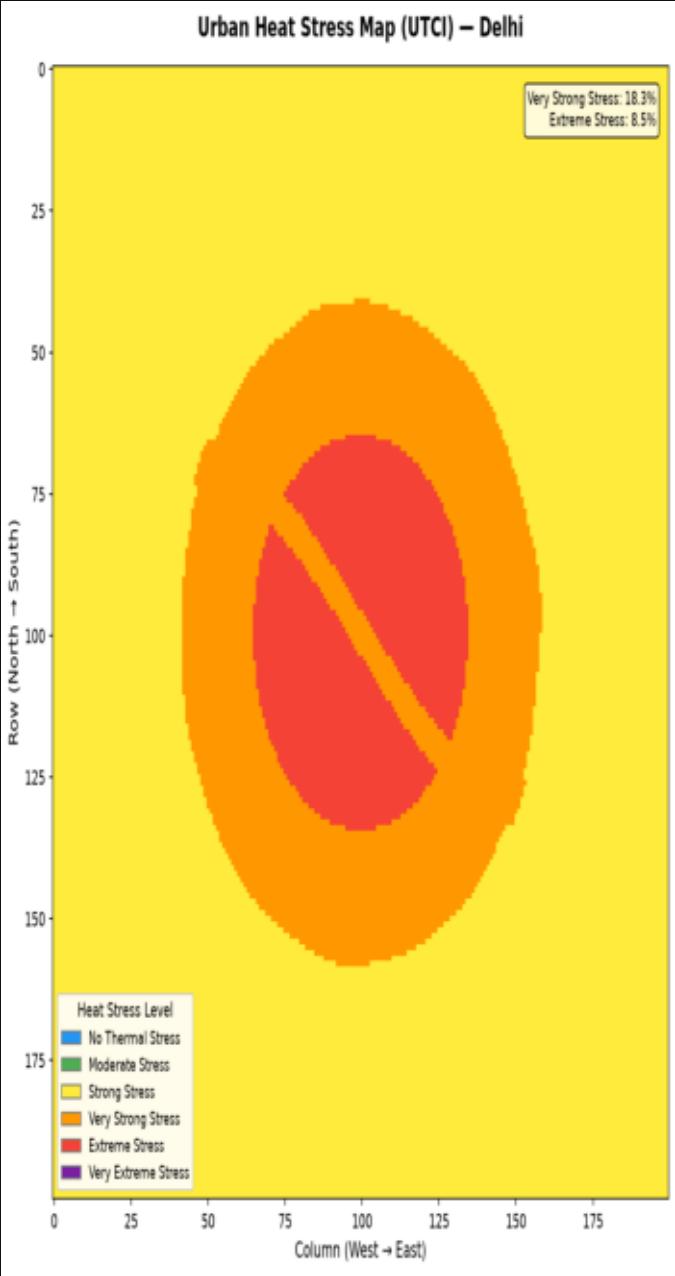
Land Use / Land Cover



Land Surface Temperature

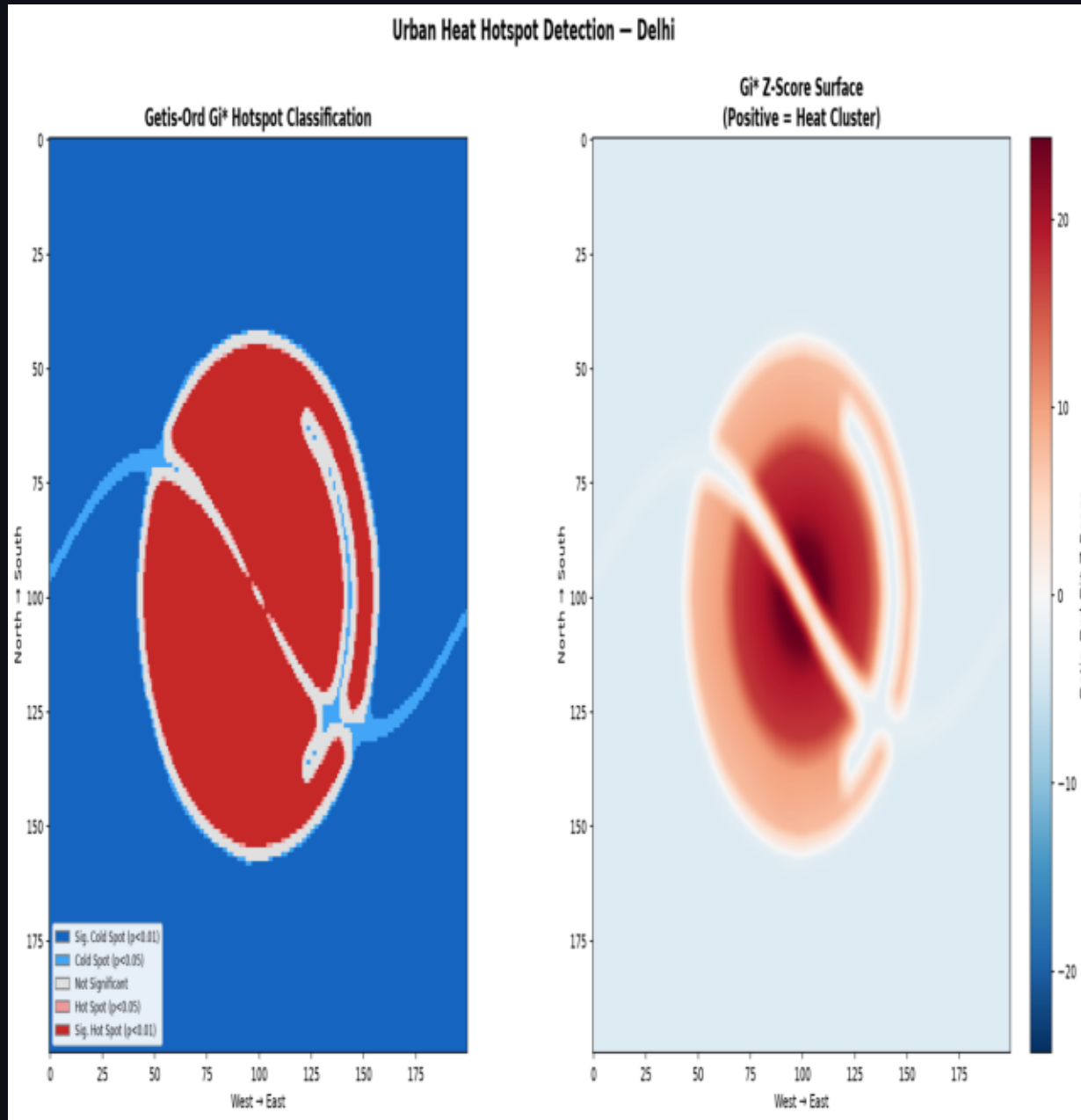


UTCI Heat Stress (6 Levels)



City: Delhi

Urban Heat Hotspot Detection – Delhi



Total Cells: 40,000

Hotspot Cells: 9,828

Hotspot Fraction: 24.57%

Extreme Hotspots: 9,447

Cold Spot Cells: 28,460

Mean LST (Hotspot): 37.88°C

Mean LST (City): 30.51°C

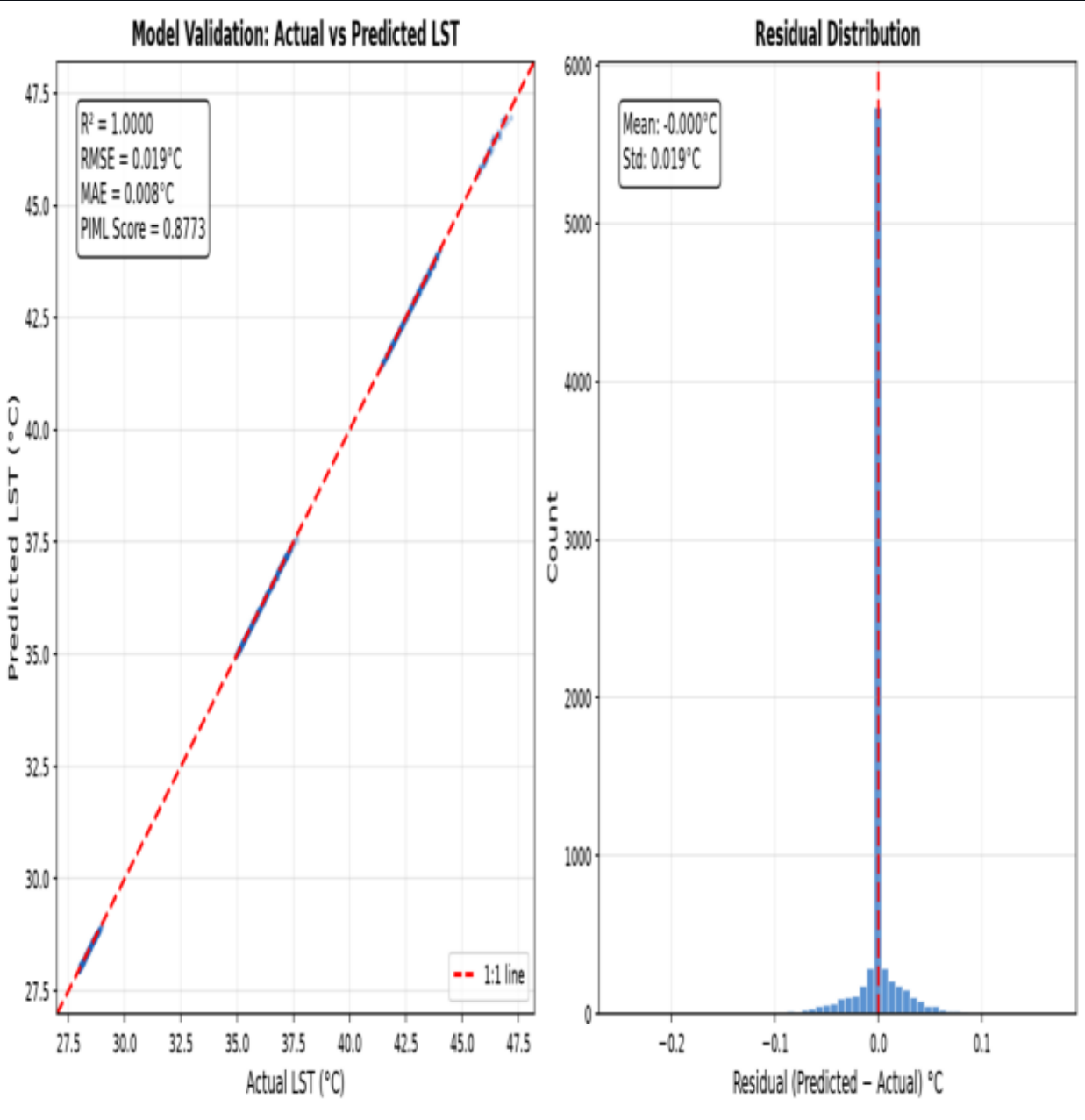
Heat Excess: +7.37°C

Max Gi* Z-score: 23.925

Physics-Informed ML Model — Validation

XGBoost + Energy Balance Constraint | R²=1.0000

Validation: Actual vs Predicted LST



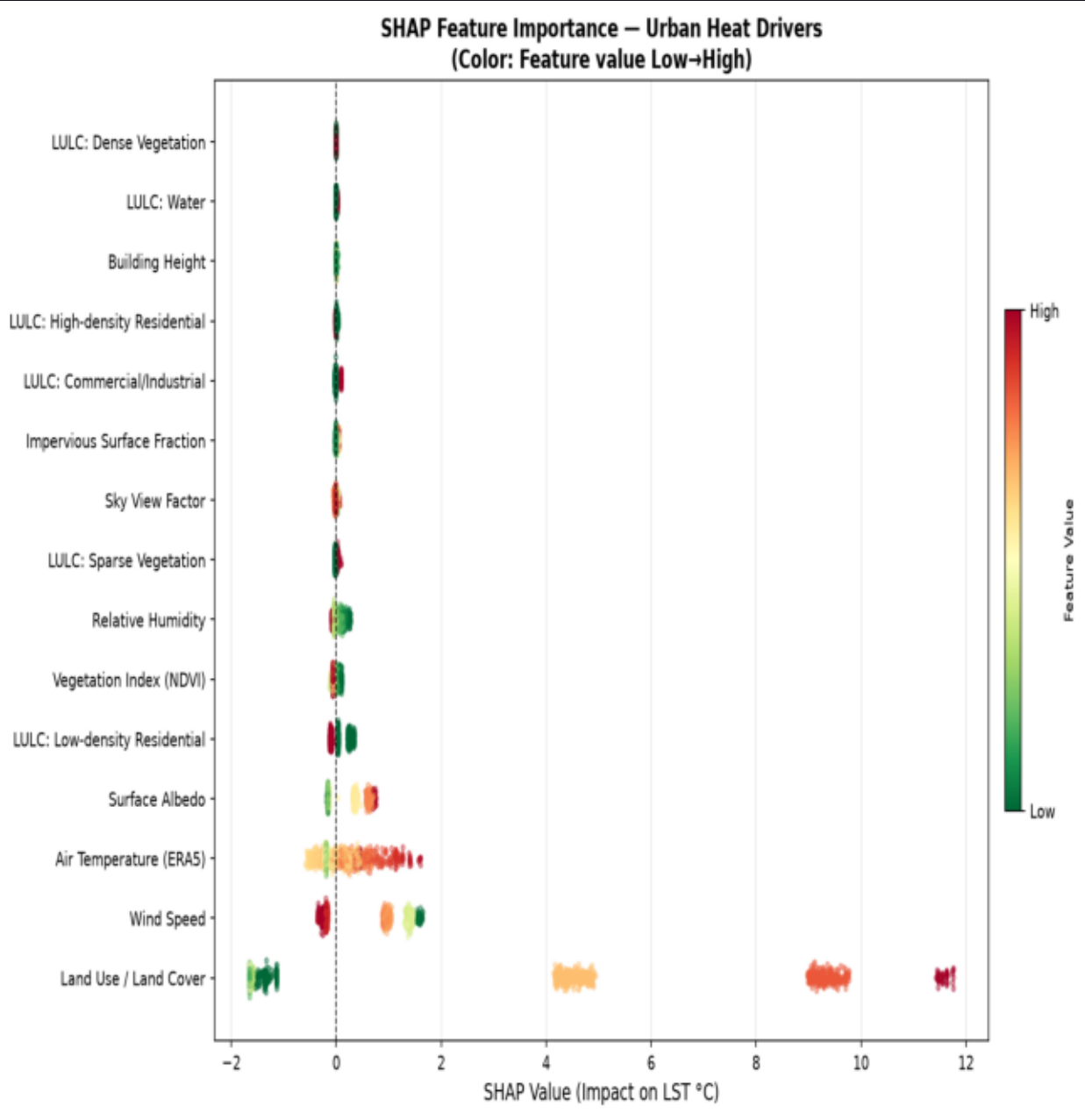
Model Metrics

R ² Score:	1.0000
RMSE:	0.019°C
MAE:	0.008°C
Physics Penalty:	0.7542
PIML Score:	0.8868
Features:	16
Train Samples:	32,000
LSTM R ² :	0.9384
LSTM RMSE:	1.599°C

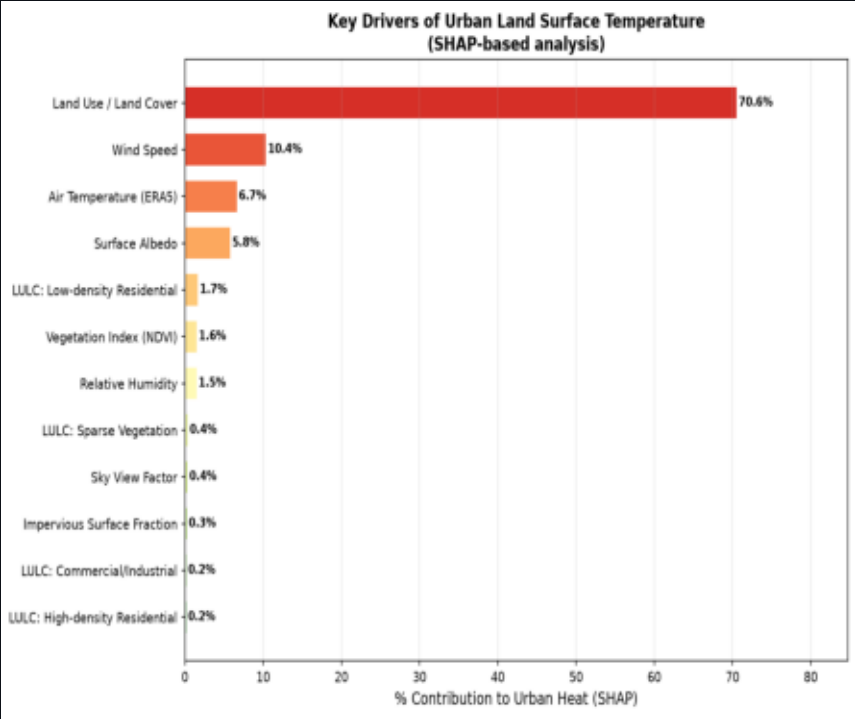
Physics Constraint

$$\begin{aligned} R_n &= H + LE + G \\ \text{Penalty} &= \lambda \times |R_n - H - LE - G|^2 \\ \lambda &= 0.15 \end{aligned}$$

SHAP Beeswarm Plot



Top Drivers (% Contribution)



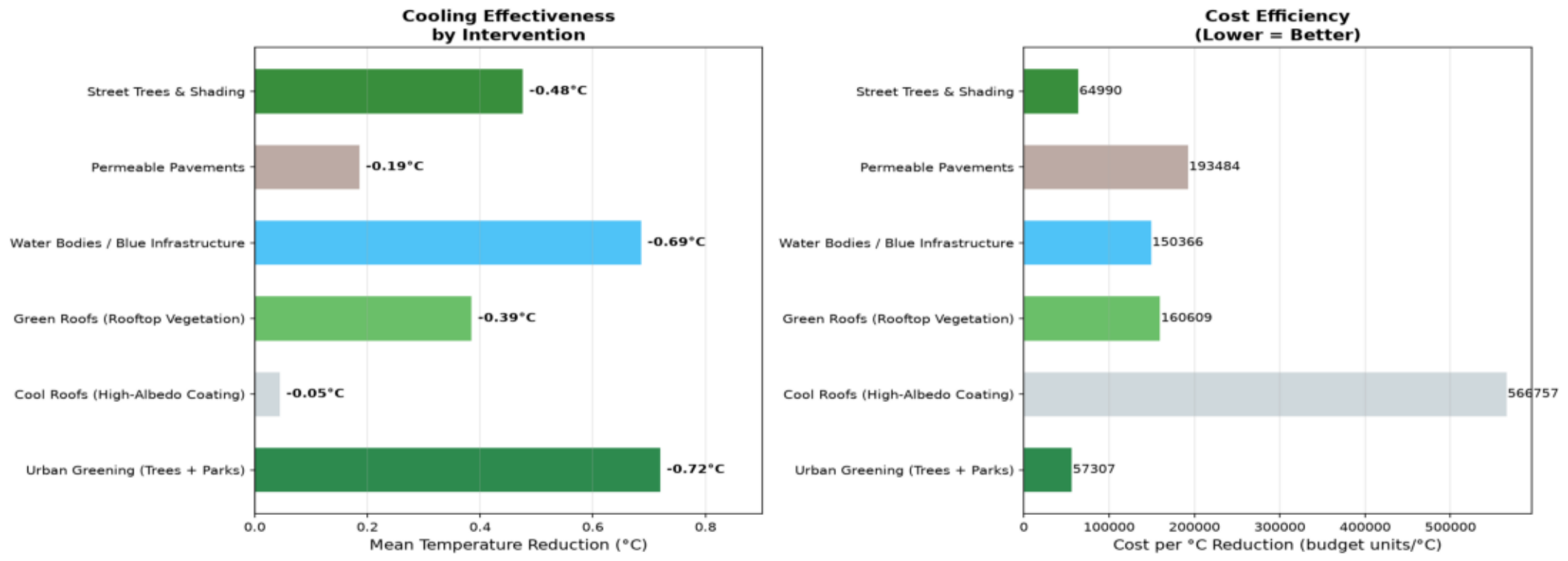
Driver Rankings

- | | |
|----------------------------------|-------|
| 1. Land Use / Land Cover | 70.6% |
| 2. Wind Speed | 10.4% |
| 3. Air Temperature (ERA5) | 6.7% |
| 4. Surface Albedo | 5.8% |
| 5. LULC: Low-density Residential | 1.7% |

6 Intervention Types | 51.8 km² Coverage Each

Cooling Effectiveness vs Cost Efficiency

Cooling Scenario Comparison — All Interventions

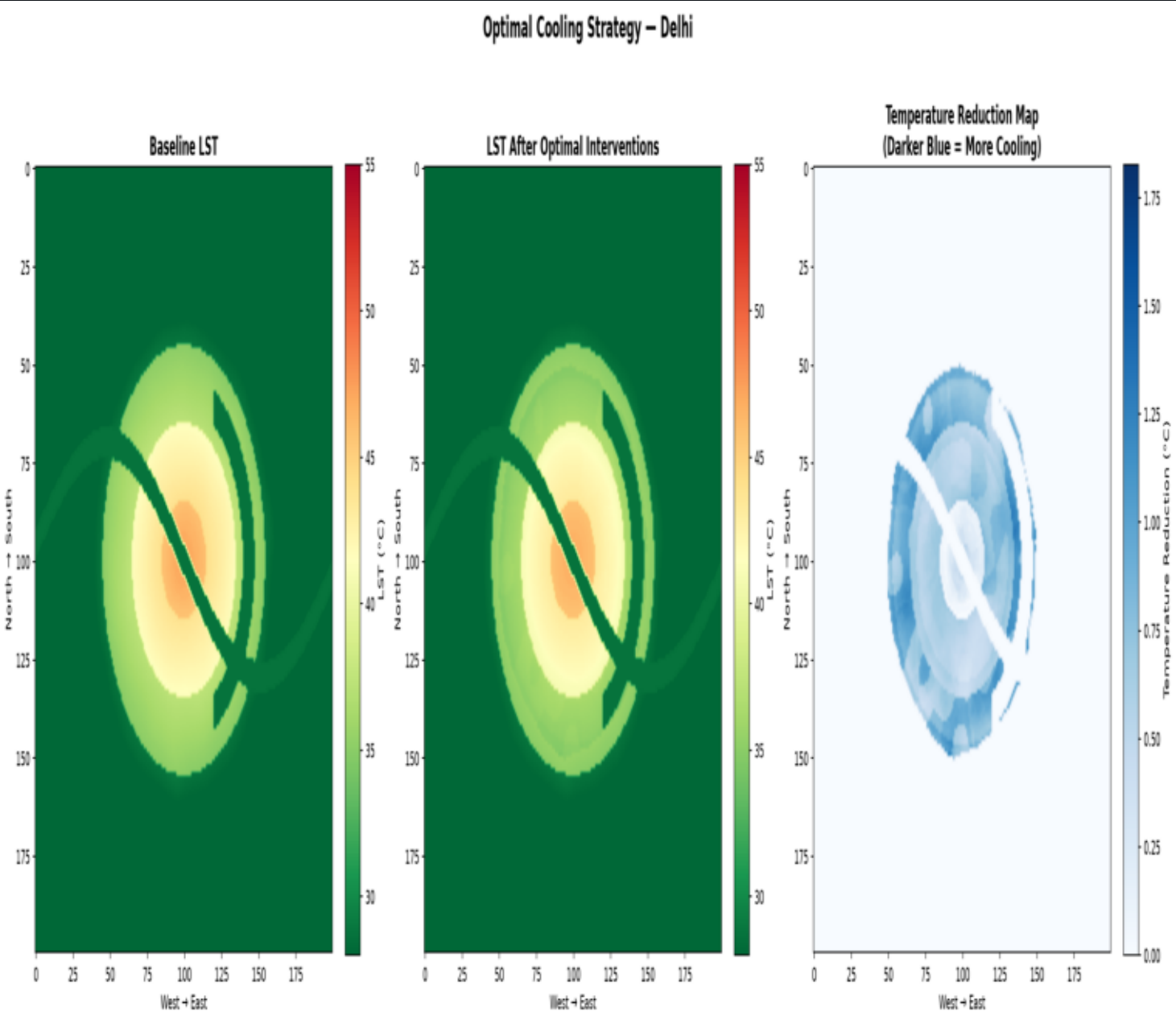


Intervention	Area (km²)	Mean ΔT (°C)	Max ΔT (°C)	Cost/°C
Urban Greening (Trees + Parks)	51.68	-0.721	-1.327	57307.3
Water Bodies / Blue Infrastructure	51.68	-0.687	-1.22	150365.8
Street Trees & Shading	51.68	-0.477	-0.922	64989.5
Green Roofs (Rooftop Vegetation)	51.68	-0.386	-0.954	160608.7
Permeable Pavements	51.68	-0.187	-0.596	193483.5
Cool Roofs (High-Albedo Coating)	51.68	-0.046	-0.29	566757.3

Optimal Cooling Strategy

Budget: 50,000 units | Area: 62.5 km² | Cooling: -0.47°C

Baseline LST | Post-Intervention LST | Temperature Reduction (ΔT)



Recommendation

Budget Used:	50,000 units
Budget Total:	50,000 units
Area Covered:	62.5 km²
Cells Treated:	6,250
Hot. Cooling:	-0.468°C
City Cooling:	-0.115°C
Intervention Mix:	
• Urban Greening	6,250

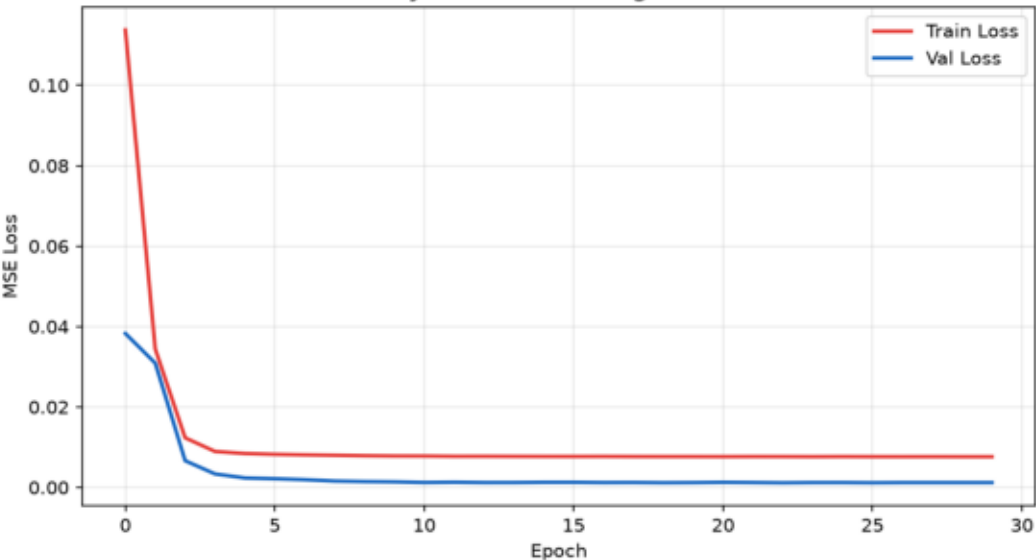
Temporal Heat Dynamics — Physics-Informed LSTM

365 days × 24h | $R^2=0.9384$ | $RMSE=1.599^{\circ}C$

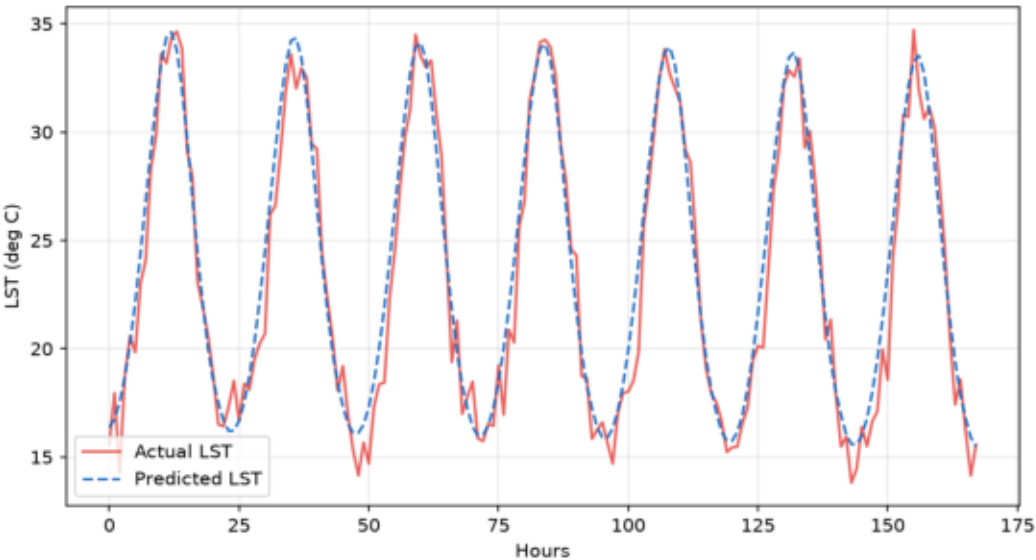
Training Curves | 7-Day Prediction | Scatter | Diurnal LULC Profiles

Physics-Informed LSTM — Urban Heat Temporal Dynamics

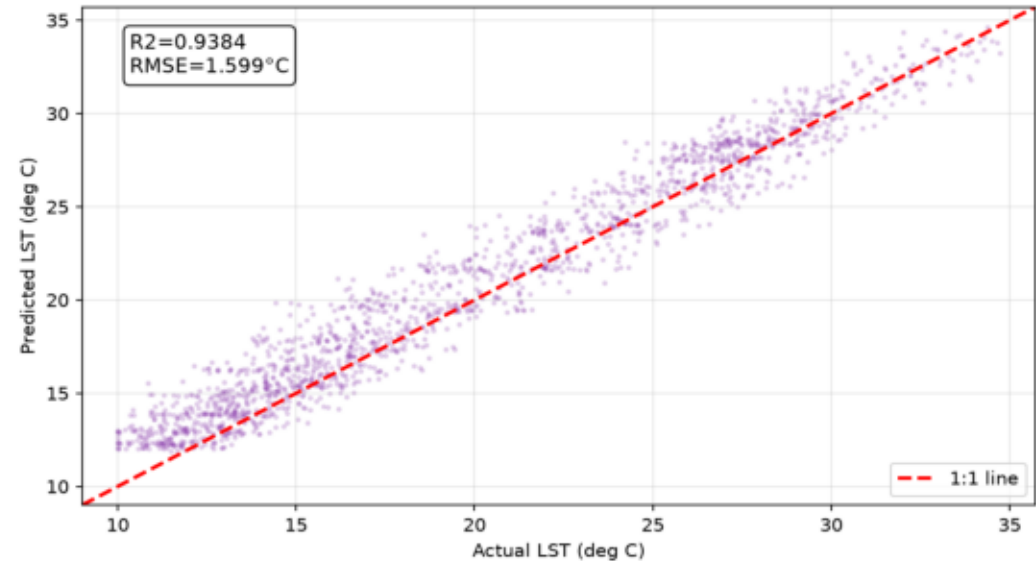
PhysicsLSTM Training Curves



7-Day Temporal Prediction
 $R^2=0.9384$ $RMSE=1.60^{\circ}C$



Actual vs Predicted LST



Diurnal LST Profile by Land Use

